Optimal Labor Income Taxation
Spring 2020

ECO 354: Public Economics

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Taxes and redistributions

Key question:

By how much should government reduce inequality using taxes and transfers?

1) Governments use **taxes** to raise revenue

2) This revenue funds **transfer** programs:

a) Universal Transfers: Public Education, Health Care Benefits (only 65+ in the US), Retirement and Disability Benefits, Unemployment benefits

b) Means-tested Transfers: In-kind (Medicaid, public housing, foodstamps in the US) and cash benefits

Modern governments raise large fraction of GDP in taxes (30-45%) and spend significant fraction of GDP on transfers

FACTS ON US TAXES AND TRANSFERS

References: Comprehensive description in:
http://www.taxpolicycenter.org/taxfacts/
A) Taxes: (1) individual income tax (fed+state), (2) payroll taxes on earnings (fed, funds Social Security+Medicare), (3) corporate income tax (fed+state), (4) sales taxes (state)+excise taxes (state+fed), (5) property taxes (state)
B) Means-tested Transfers: (1) refundable tax credits (fed), (2) in-kind transfers (fed+state): Medicaid, public housing, nutrition (SNAP), education, (3) cash welfare: TANF for single parents (fed+state), SSI for old/disabled (fed)

FEDERAL US INCOME TAX

US income tax assessed on **annual family** income (not individual)
[most other OECD countries have shifted to individual assessment]
Sum all cash income sources from family members (both from labor and capital income sources) = called **Adjusted Gross Income (AGI)**
Main exclusions: fringe benefits (health insurance, pension contributions and returns), imputed rent of homeowners, undistributed corporate profits, unrealized capital gains
$\Rightarrow$ AGI base is only 70% of national income

FEDERAL US INCOME TAX

Taxable income = AGI - deductions
deduction is max of standard deduction or itemized deductions
Standard deduction is a fixed amount ($12K for singles, $24K for married couple)
Itemized deductions: (a) state and local taxes paid (up to $10K), (b) mortgage interest payments, (c) charitable giving, various small other items
[about 10% of AGI lost through itemized deductions, called tax expenditures]

FEDERAL US INCOME TAX: TAX BRACKETS

Tax $T(z)$ is piecewise linear and continuous function of taxable income z with constant marginal tax rates (MTR) $T'(z)$ by brackets:

- In 2018+, 6 brackets with MTR 10%,12%,22%,24%,32%,35%, 37% (top bracket for z above \$600K), indexed on price inflation
- In case of India, income tax brackets are mainly only three slabs: 10%,20%, and 30%. An individual with an income above INR 10,00,000 in in the same marginal tax bracket as an individual who earns INR 1000,000,000 and above!
- Tax rates change frequently over time. Top MTRs have declined drastically since 1960s (as in many OECD countries)

Optimal Taxation: No Behavioral Responses

Utility $u(c)$ strictly increasing and concave
Same for everybody where c is after tax income.
Income z is fixed for each individual, $c = z - T(z)$ where $T(z)$ is tax/transfer on z (tax if $T(z) > 0$, transfer if $T(z) < 0$)
N individuals with fixed incomes $z_1 < ... < z_N$
Government maximizes **Utilitarian** objective:

$$SWF = \sum_{i=1}^{N} u(z_i - T(z_i))$$

subject to **budget constraint** $\sum_{i=1}^{N} T(z_i) = 0$ (taxes need to fund transfers)

No Behavioral Responses

Replace $T(z_1) = -\sum_{i=2}^{N} T(z_i)$ from budget constraint:

$$SWF = u\left(z_1 + \sum_{i=2}^{N} T(z_i)\right) + \sum_{i=2}^{N} u(z_i - T(z_i))$$

First order condition (FOC) in $T(z_j)$ for a given $j = 2, .., N$:

$$0 = \frac{\partial SWF}{\partial T(z_j)} = u'\left(z_1 + \sum_{i=2}^{N} T(z_i)\right) - u'(z_j - T(z_j)) = 0 \Rightarrow$$

$u'(z_j - T(z_j)) = u'(z_1 - T(z_1)) \Rightarrow z_j - T(z_j) =$ constant across $j = 1, .., N$

Perfect equalization of after-tax income = 100% tax rate and redistribution [draw graph]

Utilitarianism with decreasing marginal utility leads to perfect egalitarianism [Edgeworth, 1897]

Simpler Derivation: two individuals

$\max SWF = u(z_1 - T(z_1)) + u(z_2 - T(z_2))$ s.t. $T(z_1) + T(z_2) = 0$

Replace $T(z_1) = -T(z_2)$ in SWF using budget constraint:

$$SWF = u\left(z_1 + T(z_2)\right) + u(z_2 - T(z_2))$$

First order condition (FOC) in $T(z_2)$:

$$0 = \frac{dSWF}{dT(z_2)} = u'\left(z_1 + T(z_2)\right) - u'(z_2 - T(z_2)) = 0 \Rightarrow$$

$u'(z_1 + T(z_2)) = u'(z_2 - T(z_2)) \Rightarrow u'(z_1 - T(z_1)) = u'(z_2 - T(z_2))$
$\Rightarrow z_1 - T(z_1) = z_2 - T(z_2)$ constant across the 2 individuals
Perfect equalization of after-tax income = 100% tax rate and redistribution [see graph]

ISSUES WITH SIMPLE MODEL

1) **No behavioral responses:** Obvious missing piece: 100% redistribution would destroy incentives to work and thus the assumption that z is exogenous is unrealistic
$\Rightarrow$ Optimal income tax theory incorporates behavioral responses
2) **Issue with Utilitarianism:** Even absent behavioral responses, many people would object to 100% redistribution [perceived as confiscatory]
$\Rightarrow$ Citizens' views on fairness impose **bounds** on redistribution govt can do [political economy / public choice theory]

EQUITY-EFFICIENCY TRADE-OFF

Taxes can be used to raise revenue for transfer programs which can reduce inequality in disposable income

$\Rightarrow$ Desirable if society feels that inequality is too large

Taxes (and transfers) reduce incentives to work

$\Rightarrow$ High tax rates create economic inefficiency if individuals respond to taxes

Size of behavioral response limits the ability of government to redistribute with taxes/transfers

$\Rightarrow$ Generates an equity-efficiency trade-off

Empirical tax literature estimates the size of behavioral responses to taxation

Labor Supply Theory

Individual has utility over labor supply l and consumption c: $u(c, l)$ **increasing** in c and **decreasing** in l [= increasing in leisure]

$$\max_{c,l} u(c, l) \quad \text{subject to} \quad c = w \cdot l + R$$

with $w = \bar{w} \cdot (1 - \tau)$ the net-of-tax wage ($\bar{w}$ is before tax wage rate and τ is tax rate), and R non-labor income
FOC $w\frac{\partial u}{\partial c} + \frac{\partial u}{\partial l} = 0$ defines Marshallian labor supply $l = l(w, R)$

Uncompensated labor supply elasticity: $\varepsilon^u = \dfrac{w}{l} \cdot \dfrac{\partial l}{\partial w}$

Income effects: $\eta = w\dfrac{\partial l}{\partial R} \leq 0$ (if leisure is a normal good)

Labor supply curve

- an individual is endowed with a T amount of time which she can either utilize on working or spending it the way she pleases- *leisure (l)*. These two means of spending time has the following competing effects.
- An hour spent working earns the individual a wage at the rate ω- dependent on her human capital and training- but we treat those exogenous here. However, an hour spent working means that hour taken away from her *leisure*- which is the opportunity cost of working for an individual.
- Utility of an individual depends on her consumption- funded from her wage income, and on her leisure. Wage rate earned is treated as a constant for an individual but might differ across agents. Let is assume R is the no labour income of an individual.
- The time allocation problem for an agent:

$$\max_{c,l} \quad u(c.l) \quad \text{subject to} \quad pc \leq \omega(T - l) + R$$

Labor supply

- The optimization problem can be rewritten as:

$$\max_{c,l} \quad u(c.l) \quad \text{subject to} \quad pc + \omega l \leq wT + R \tag{1}$$

- The FOCs of an interior maximum (guaranteed by concavity of $u(.)$):

$$u_c = \lambda p \tag{2}$$
$$u_l = \lambda \omega \tag{3}$$

- The optimal allocation of time is given by:

$$\frac{u_c}{u_l} = \frac{p}{\omega} \tag{4}$$
$$pc + \omega l = \omega T + R \tag{5}$$

Graph space

Equilibrium

- The optimal labor supply from an individual occurs when the indifference curves (level sets of the utility function $u(c, l)$) is tangent to the budget constraint- both drawn on a (c, l) space. The consumption and labour supply can be seen from the equilibrium intercepts.
- Optimal labour supply therefore is a function of wage rate, consumption prices and total time endowment.
- Explanation: Time allocation problem- where an unit of time spent working earns w and increases consumption by $\frac{w}{p}$ at the expense of a forgone utility from leisure consumption $u_l(c, l)$. An equilibrium, therefore, is defines as a work-leisure consumption balance where no utility increasing reshuffling exists.
- **Equilibrium:** An interior equilibrium happens at a point where an unit more of leisure consumption and consequent forgone consumption exactly matches in terms of utility.

Comparative statics

- **Question** what happens to an equilibrium labour supply decision when wage rate goes down? This is a very relevant question because a labour income taxation reduces the per hour wage that an individual receives by supplying her labour.
- Does a labour income tax or an decrease in wage certainly reduces the labour supply? By how much?
- **counteractive forces** A change in wage beings two counteractive forces that decides the labour supply.
  - **subs. effct:** an reduction in wage makes leisure *cheaper*, leading to a larger leisure consumption for everyone. leading to a fall in labour supply.
  - **income effct:** a reduction in wage also reduces the income of the individual and if leisure is a normal good, it leads to a reduction in leisure consumption as well, leading to an increase in labour supply.

  The final effect depends on the relative strengths of these effects.

Labour supply theory

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Theory continued...

Substitution effects: Hicksian labor supply: $l^c(w, u)$ minimizes cost needed to reach u given slope $w \Rightarrow$

$$\text{Compensated elasticity} \quad \varepsilon^c = \frac{w}{l} \cdot \frac{\partial l^c}{\partial w} > 0$$

$$\text{Slutsky equation} \quad \frac{\partial l}{\partial w} = \frac{\partial l^c}{\partial w} + l\frac{\partial l}{\partial R} \Rightarrow \varepsilon^u = \varepsilon^c + \eta$$

Tax rate τ discourages work through substitution effects (work pays less at the margin)
Tax rate τ encourages work through income effects (taxes make you poorer and hence in more need of income)
Net effect ambiguous (captured by sign of ε^u)

Graph space

General non-linear income tax

With no taxes: $c = z$ (consumption = earnings)
With taxes $c = z - T(z)$ (consumption = earnings - net taxes)
$T(z) \geq 0$ if individual pays taxes on net, $T(z) \leq 0$ if individual receives transfers on net
$T'(z) > 0$ reduces net wage rate and reduces labor supply through substitution effects
$T(z) > 0$ reduces disposable income and increases labor supply through income effects
$T(z) < 0$ increases disposable income and decreases labor supply through income effects
Transfer program such that $T(z) < 0$ and $T'(z) > 0$ always discourages labor supply

Optimal linear tax rate: Laffer curve

$c = (1-\tau) \cdot z + R$ with τ linear tax rate and R fixed universal transfer funded by taxes $R = \tau \cdot Z$ with Z average earnings
Individual $i = 1, .., N$ chooses l_i to max $u^i((1-\tau) \cdot w_i l_i + R, l_i)$
Labor supply choices l_i determine individual earnings $z_i = w_i l_i \Rightarrow$
Average earnings $Z = \sum_i z_i / N$ depends (positively) on net-of-tax rate $1-\tau$.
Tax Revenue per person $R(\tau) = \tau \cdot Z(1-\tau)$ is inversely U-shaped with τ: $R(\tau = 0) = 0$ (no taxes) and $R(\tau = 1) = 0$ (nobody works): called the Laffer Curve

Graph space

Optimal linear tax rate: Laffer curve

Top of the Laffer Curve is at τ^* maximizing tax revenue:

$$0 = R'(\tau^*) = Z - \tau^* \frac{dZ}{d(1-\tau)} \Rightarrow \frac{\tau^*}{1-\tau^*} \cdot \frac{1-\tau^*}{Z} \frac{dZ}{d(1-\tau)} = 1$$

Revenue maximizing tax rate: $\tau^* = \dfrac{1}{1+e}$ with $e = \dfrac{1-\tau}{Z} \dfrac{dZ}{d(1-\tau)}$

e is the elasticity of average income Z with respect to the net-of-tax rate $1-\tau$ [empirically estimable]

Inefficient to have $\tau > \tau^*$ because decreasing τ would make taxpayers better off (they pay less taxes) and would increase tax revenue for the government [and hence univ. transfer R]

If government is **Rawlsian** (maximizes welfare of the worst-off person with no earnings) then $\tau^* = 1/(1+e)$ is optimal to make transfer $R(\tau)$ as large as possible

Optimal linear tax rate: formula

Government chooses τ to maximize **utilitarian** social welfare

$$SWF = \sum_i u^i((1-\tau)w_i l_i + \tau \cdot Z(1-\tau), l_i)$$

taking into account that labor supply l_i responds to taxation and hence that this affects the tax revenue per person $\tau \cdot Z(1-\tau)$ that is redistributed back as transfer to everybody

Government first order condition: (using the envelope theorem as l_i maximizes u^i):

$$0 = \frac{dSWF}{d\tau} = \sum_i \frac{\partial u^i}{\partial c} \cdot \left[-z_i + Z - \tau \frac{dZ}{d(1-\tau)} \right],$$

Optimal linear tax rate: formula

Hence, we have the following optimal linear income tax formula

$$\tau = \frac{1-\bar{g}}{1-\bar{g}+e} \quad \text{with} \quad \bar{g} = \frac{\sum_i z_i \cdot \frac{\partial u^i}{\partial c}}{Z \cdot \sum_i \frac{\partial u^i}{\partial c}}$$

$0 \leq \bar{g} < 1$ as $\frac{\partial u^i}{\partial c}$ lower when income z_i is high (marginal utility falls with consumption)
τ decreases with elasticity e [efficiency] and with $\bar{g}$ [equity]
Formula captures the **equity-efficiency trade-off**
$\bar{g}$ is low and τ close to Laffer rate $\tau^* = 1/(1+e)$ when
(a) inequality is high
(b) marginal utility decreases fast with income